

Elyes Darouich

AI & Full-Stack Engineer

Bizerte / Tunis, Tunisia | +216 94 906 400 | elyes.darouich@esprit.tn
github.com/ElyesDarouich | github.com/ElyesD1 | linkedin.com/in/elyes-darouich | Portfolio

PROFESSIONAL SUMMARY

AI engineer specializing in agentic systems, multi-agent orchestration, and production RAG/LLM pipelines on a strong full-stack foundation. Final-year Computer Engineering student delivering an end-of-studies AI ISO-compliance platform at Talan, built on guardrailed retrieval over Llama 3.3 70B with clean-architecture FastAPI services.

TECHNICAL SKILLS

AI / ML: LangGraph, Multi-Agent Orchestration, RAG, MCP, Claude Agent SDK, Prompt Engineering, LLM Guardrails, Vector Search (ChromaDB, Qdrant), Embeddings

Languages: Python, TypeScript, JavaScript, Java, C++, C, Swift, Kotlin, Dart, Solidity, SQL

Frontend & Mobile: React, Next.js, Flutter, SwiftUI, Kotlin, Tailwind CSS

Backend & Data: FastAPI, NestJS, Flask, Node.js, Spring Boot, Celery, REST, WebSockets, PostgreSQL, MongoDB, Redis

DevOps & Web3: Docker, GitHub Actions CI/CD, Linux, Nginx, Vercel, Zero-Knowledge (Groth16, Circom, snarkjs), Hyperledger Besu, Solidity

EXPERIENCE

End-of-Studies AI Engineer (PFE) — Talan Tunisia

Feb 2026 – Jun 2026

Tunis, Tunisia

- Architected “SMI Virtual Auditor,” an AI ISO-compliance audit platform producing clause-by-clause verdicts with cited evidence (ISO 9001/14001/27001, 110 clauses) via a 3-pass RAG pipeline on Llama 3.3 70B (Groq, GitHub Models, NVIDIA NIM).
- Engineered a 7-layer guardrail system (groundedness, fidelity, contradiction, confidence capping) that suppresses hallucinations, grounding verdicts via local bge-m3 embeddings on Ollama + ChromaDB.
- Built a clean-architecture FastAPI backend (JWT RS256, 4-role RBAC, Celery, PostgreSQL/Redis/MinIO, Dockerized, 108 tests) and a React 18 + TypeScript + Zustand SPA with streaming RAG chat (SSE), PDF export, and bilingual FR/EN.

Software Engineering Intern — Talan Tunisia

Jul 2025 – Sep 2025

Tunis, Tunisia

- Built “ProjectFlow,” a cross-platform project-management platform (Flutter + NestJS) sustaining sub-150ms API responses for 100+ concurrent users at 78% test coverage, with real-time Kanban and chat over WebSockets/Socket.IO.
- Integrated an AI assistant on Google Gemini 2.0 Flash and JWT + Google OAuth with RBAC, shipped via Dockerized GitHub Actions CI/CD.

Flutter Developer Intern — Vatech WABAG

Summer 2023

Tunis, Tunisia

- Built a cross-platform employee leave-management app on a REST API with secure auth, RBAC, and Firebase real-time push notifications keeping approval workflows in sync across devices.

PROJECTS

SentinelAI

LangGraph, FastAPI, Celery, Redis, Qdrant

Production multi-agent platform scoring geopolitical & market risk for Gold/Oil/S&P 500/BTC/ETH — a 7-agent LangGraph pipeline with Monte Carlo + GARCH risk, sub-30s end-to-end at \$0 LLM cost via free-tier key rotation.

ZK-Attest

Circom, snarkjs, Solidity, Hyperledger Besu, Next.js

Privacy-preserving credential verification proving facts without revealing data — Groth16 zero-knowledge proofs, W3C Verifiable Credentials, and an ERC-5192 soulbound trust chain on Hyperledger Besu (GDPR by design).

Agentic Valley

Claude Agent SDK, TypeScript, Node.js, VSCode API

VSCode extension visualizing multi-agent orchestration as a pixel-art office — a main agent dispatches 7 specialized Claude Agent SDK agents via the Task tool, with reducer-driven sprite states.

EDUCATION

National Engineering Degree, Computer Engineering — ESPRIT, Tunis

2021 – 2026

Baccalaureate in Computer Science

2021

CERTIFICATIONS & LANGUAGES

Certifications: AWS Cloud Foundations • NVIDIA Generative AI with Diffusion Models • Hedera Hashgraph Developer (2025)

Languages: Arabic (Native) • English (C1) • French (B2)